SPINN Paul-Trap PINNs

Hyperparameters, derivative validation, and coordinate projection

- Runs completed: 19.
- Best DEVSIM match: bc_weight_1000 with Phi MSE 0.619.
- Best fixed model passes the single-seed derivative-aware validation gate.
- Coordinate projection is implemented and operator-tested; converged geometry optimization is still pending.

Study Setup

- Ground truth: Circular Trap DEVSIM RF solve with East/West electrodes at 300 V.
- PINN target: solve the RF potential on matching geometry.
- Comparison: evaluate the trained PINN at DEVSIM mesh nodes and compute common-point error metrics.
- Sweeps: width, depth, activation, boundary-condition weight, optimizer, and learning rate.

Governing Equation and PINN Objective

The RF potential is modeled as a Laplace problem:

$$\nabla^2 \phi_{\mathrm{RF}}(x, y) = 0,$$

with electrode boundary conditions enforced through a weighted boundary loss:

$$\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{BC}} \mathcal{L}_{\mathrm{BC}},$$

$$\mathcal{L}_{\mathrm{PDE}} = \frac{1}{N_c} \sum_i \left| \nabla^2 \phi_\theta(x_i, y_i) \right|^2, \qquad \mathcal{L}_{\mathrm{BC}} = \frac{1}{N_b} \sum_j \left| \phi_\theta(x_j, y_j) - \phi_j \right|^2.$$

Evaluation Metrics

Training curves were compared using normalized loss:

$$\widehat{\mathcal{L}}(t) = \frac{\mathcal{L}(t)}{\mathcal{L}(0)}.$$

Ground-truth agreement was evaluated with DEVSIM common-node mean-square error:

$$\mathrm{MSE}_{\phi} = \frac{1}{N}\sum_{i}\left(\phi_{\theta,i} - \phi_{\mathrm{DEVSIM},i}\right)^2.$$

The RF pseudopotential was computed from the RF-field gradient:

$$\Psi_{\mathrm{RF}} = \frac{q^2}{4m\Omega^2}\left|\nabla\phi_{\mathrm{RF}}\right|^2, \qquad \mathrm{MSE}_{\Psi} = \frac{1}{N}\sum_{i}\left(\Psi_{\theta,i} - \Psi_{\mathrm{DEVSIM},i}\right)^2.$$

Main Results

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
bc_weight_1000	0.00126	2.09e+03	0.619	0.0388
width_256	0.0685	2.78e+03	10.4	0.955
activation_sinusoidal	0.0089	3.03e+03	12.3	1.75
optimizer_rmsprop	0.0145	1.94e+03	14.5	2.38
optimizer_adamw	0.0103	3.63e+03	21.9	3.33
baseline	0.0188	3.48e+03	58.9	6.89
width_64	0.02	5.37e+03	69.2	7.33
lr_3e-3	0.493	3.43e+03	142	6.57

- Best DEVSIM agreement came from bc_weight_1000: Phi MSE 0.619, Psi MSE 0.0388.
- That is about 95x lower RF-potential MSE than the baseline.
- ReLU is the cautionary case: tiny training loss but the worst DEVSIM field error.
- Among optimizer-only variants, RMSprop was best: Phi MSE 14.5.

Derivative-Aware Audit of the Best Fixed Model

Quantity	Audited error	Proposed gate
Potential MAE / 300 V	0.222%	$< 0.5\%$
Electric-field relative L^2	1.33%	$< 5\%$
Center Hessian relative Frobenius	0.77%	$< 5\%$
Quadrupole-efficiency relative error	0.23%	$< 5\%$
Secular-frequency relative L^2	0.72%	$< 5\%$

- Single-seed derivative gate: **PASS**.
- This certifies a fixed-geometry checkpoint, not robustness across seeds or geometries.
- The earlier width-64 diagnostic was not the best model and should not represent the final baseline.

Mentor-Specified Residual Projection

The mentor-specified shape network follows Equation 13 of Zhang et al.:

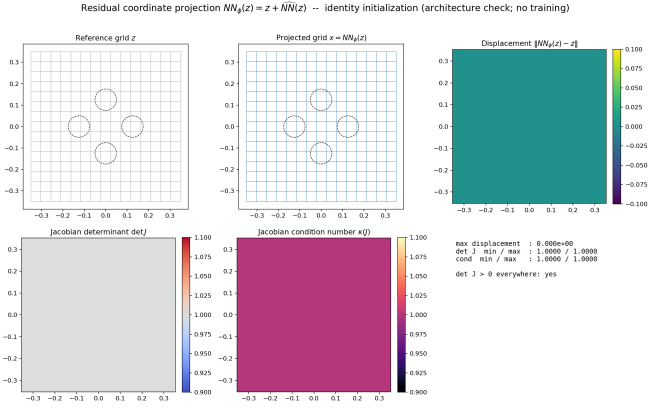
$$NN_{\phi}(\mathbf{z}) = \mathbf{z} + \widehat{NN}_{\phi}(\mathbf{z}), \qquad \widehat{NN}_{\phi}(\mathbf{z})\big|_{\text{init}} = \mathbf{0}.$$

The physical-field network is evaluated on the projected coordinates:

$$\mathbf{x} = NN_{\phi}(\mathbf{z}), \qquad \phi_{\theta}(\mathbf{x}) = NN_{\theta}(NN_{\phi}(\mathbf{z})).$$

- Both networks use six hidden layers of width 50 with tanh in the reference method.
- The residual output layer is zero-initialized, making the complete shape map exactly identity initially.
- Positive-Jacobian and fixed-region penalties preserve topology and anchor non-design regions.
- Shape and field parameters are ultimately trained jointly against physics, constraints, and the trap objective.

Residual Projection Architecture Diagnostic



- Diagnostic of the Equation-13 residual map $NN_\phi(\mathbf{z}) = \mathbf{z} + \widehat{NN}(\mathbf{z})$ on the $[-0.35, 0.35]^2$ scan.
- Shown at identity initialization: displacement ≈ 0, $\det J \approx 1$, condition number ≈ 1.
- This is an architecture check for the freeform Stage 2 network, not a trained geometry.
- No optimized freeform geometry is claimed; a non-trivial map only appears after joint training (or an

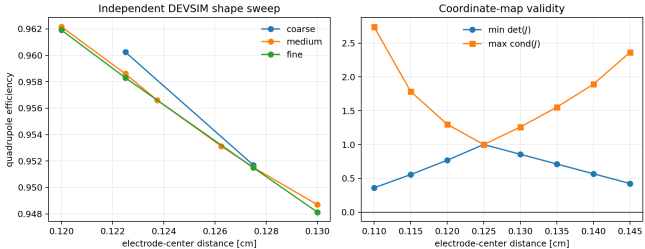
Constrained Map vs. Freeform Network

For the constrained validation map $\mathbf{x} = F_d(\mathbf{z})$, the equivalent pulled-back PDE is

$$\Delta_{\mathbf{x}}\phi = \frac{1}{\det J}\nabla_{\mathbf{z}} \cdot \left(\det J\, J^{-1} J^{-T} \nabla_{\mathbf{z}} u_\theta\right) = 0.$$

- Stage 1 uses an analytic electrode-translation map to validate geometry derivatives against DEVSIM.
- Stage 2 replaces that analytic map with the trainable residual network from Equation 13.
- The analytic map is a controlled benchmark, not the intended final shape architecture.
- Ordinary $\Delta_{\mathbf{z}} u_\theta = 0$ would still solve the wrong transformed problem.

Coordinate Map and Independent Solver Audit



- Dense map scan: minimum $\det J = 0.359$.
- Maximum Jacobian condition number: 2.74.
- Four manufactured coordinate-map tests recover harmonic solutions.
- The full coordinate-projection model has only been smoke-tested; no optimized geometry is claimed.

DEVSIM Shape-Gradient Convergence

DEVSIM mesh	3-point derivative	5-point derivative	stencil spread
coarse	-1.7100	–	–
medium	-1.4164	-1.4396	1.61%
fine	-1.3618	-1.3565	0.39%

- Fine-mesh three- and five-point derivatives agree within 0.39%.
- Fine vs. medium discretization changes the derivative by about 6%; this is the present reference uncertainty.
- Local DEVSIM efficiency changes by only 0.0045% across fit radii 0.015–0.06 cm.
- A PINN shape derivative is accepted only after agreement with this independent reference.

What Is a Result Today?

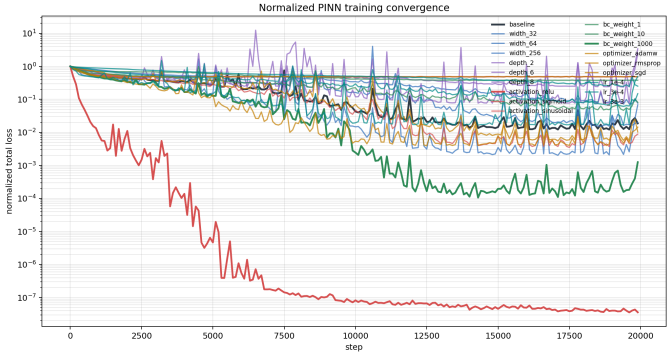
Validated now

- Best fixed-geometry checkpoint passes derivative-aware gates.
- Pullback operator passes affine and radial manufactured tests.
- Coordinate map is injective over the tested one-parameter interval.
- Independent DEVSIM shape-gradient workflow is operational.

Not yet a scientific result

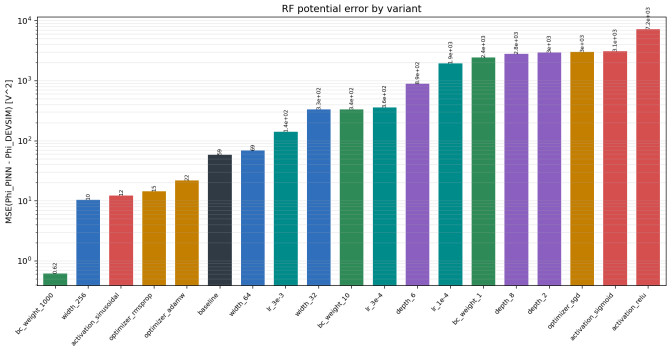
- Three-seed confirmation of the best fixed model.
- Converged training of $u_\theta(\mathbf{z}, d)$.
- PINN–DEVSIM shape-gradient agreement after convergence.
- A physical 3D escape depth or a freeform optimized electrode.

Normalized Convergence



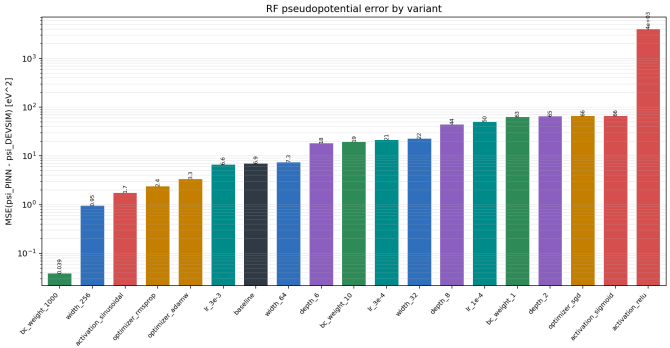
- Normalizing by initial loss makes runs with different starting scales comparable.
- Fast normalized convergence does not guarantee physical accuracy.
- ReLU ended at normalized loss 3.58e-08, but failed the DEVSIM comparison.
- BC weight 1000 had both low normalized loss and low field error: Phi MSE 0.619.

RF Potential Error



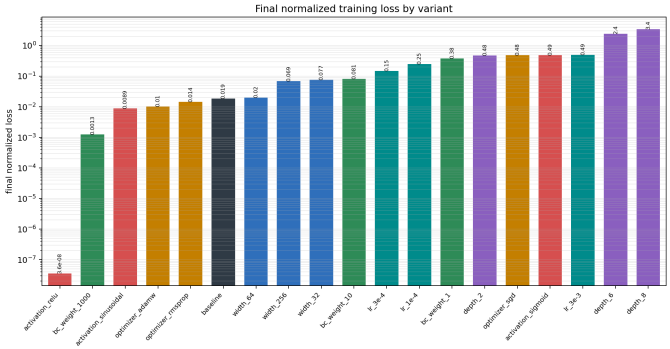
- Lowest RF-potential error: bc_weight_1000 at Phi MSE 0.619.
- Largest RF-potential error: activation_relu at Phi MSE 7.2e+03.
- Boundary-condition weighting was the strongest lever in this sweep.
- Wider networks and sinusoidal activation were strong architecture-side options.

RF Pseudopotential Error



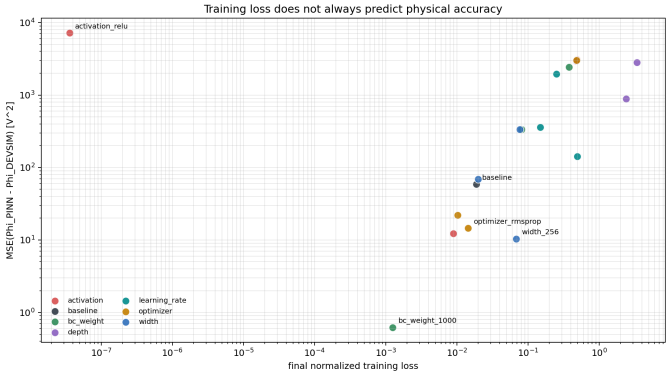
- Lowest pseudopotential error: bc_weight_1000 at Psi MSE 0.0388.
- The pseudopotential depends on gradients, so it exposes field-shape errors.
- The best Phi runs generally also improved Psi, but noisy PDE spikes can still damage gradients.

Final Normalized Loss



- Final training loss alone is not a reliable model-selection metric.
- ReLU is artificially favored because its second derivative is zero almost everywhere.
- BC weight 1000 is the most credible winner because it combines low loss with low DEVSIM error.

Training Loss vs. Field Error



- Ideal runs appear near the lower-left: low training loss and low DEVSIM error.
- ReLU is isolated as a false low-loss solution.
- BC weight 1000 is the cleanest point because it pairs low loss with the lowest field error.

Baseline Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
baseline	0.0188	3.48e+03	58.9	6.89

- Baseline Phi MSE is 58.9; baseline Psi MSE is 6.89.
- Several variants beat this reference decisively, so the PINN is hyperparameter-sensitive.

Width Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
width_256	0.0685	2.78e+03	10.4	0.955
width_64	0.02	5.37e+03	69.2	7.33
width_32	0.0767	7.27e+03	334	22.5

- Best width run: width_256 with Phi MSE 10.4.
- Worst width run: width_32 with Phi MSE 334.
- Width improved physical field accuracy more than final loss alone suggested.

Depth Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
depth_6	2.4	7.87e+03	886	18.3
depth_8	3.41	1.26e+04	2.81e+03	44.4
depth_2	0.478	9.79e+03	2.95e+03	65.4

- Best depth run: depth_6 with Phi MSE 886.
- Depth variants were unstable; increasing depth did not improve DEVSIM agreement.

Activation Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
activation_sinusoidal	0.0089	3.03e+03	12.3	1.75
activation_sigmoid	0.486	1.01e+04	3.1e+03	65.6
activation_relu	3.58e-08	170	7.2e+03	3.97e+03

- Sinusoidal activation was the best activation-only variant.
- ReLU produced misleadingly tiny loss because the Laplace residual degenerates under second derivatives.

BC Weight Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
bc_weight_1000	0.00126	2.09e+03	0.619	0.0388
bc_weight_10	0.0814	5.28e+03	335	19.3
bc_weight_1	0.379	9.75e+03	2.44e+03	63.1

- Best boundary-weight run: bc_weight_1000 with Phi MSE 0.619.
- Too little boundary weight left boundary mismatch dominant; 1000 gave the cleanest DEVSIM match.

Optimizer Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
optimizer_rmsprop	0.0145	1.94e+03	14.5	2.38
optimizer_adamw	0.0103	3.63e+03	21.9	3.33
optimizer_sgd	0.481	1.01e+04	3.02e+03	65.5

- Best optimizer-only run: optimizer_rmsprop with Phi MSE 14.5.
- SGD plateaued with poor boundary accuracy; RMSprop was the strongest optimizer-only result.

Learning Rate Sweep

Variant	Final norm. loss	AUC norm. loss	Phi MSE	Psi MSE
lr_3e-3	0.493	3.43e+03	142	6.57
lr_3e-4	0.149	6.49e+03	358	21.3
lr_1e-4	0.251	1.05e+04	1.95e+03	50.2

- Best learning-rate run: lr_3e-3 with Phi MSE 142.
- Lower learning rates were smoother but too slow for the fixed training budget.

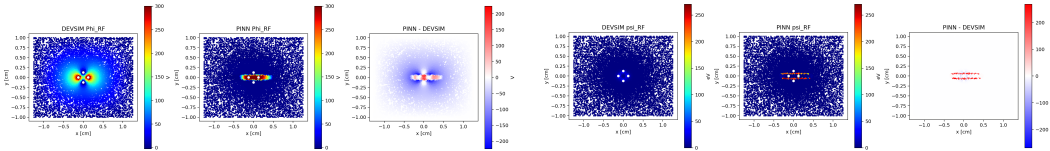
Discussion Points

- Repeat BC weight 1000, width 256, and sinusoidal activation across at least three seeds.
- Should ReLU be excluded for second-derivative PINNs, or kept only as a negative control?
- Train the parametric coordinate-projection model until its DEVSIM shape derivative passes the reference gate.
- After the one-parameter proof, compare constrained freeform maps with PIXEL, PIG, and harmonic-by-construction backends.

Error Map Appendix

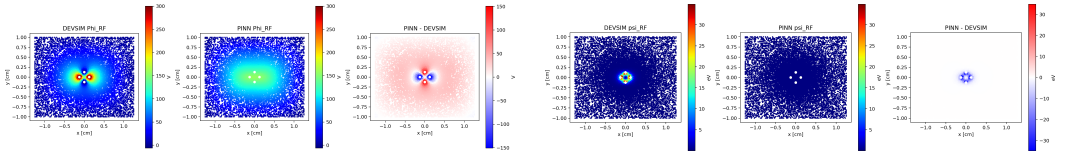
- Each following slide shows raw PINN minus DEVSIM error maps.
- Left: RF potential error.
- Right: RF pseudopotential error.
- Text remains black on a transparent/plain slide background; plots are included as generated image data.

activation_relu



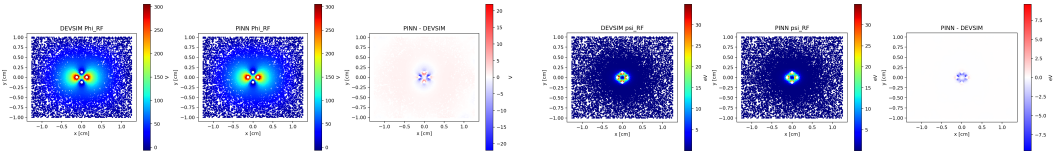
Misleading loss case: training loss collapses, but field error is the worst in the sweep.

activation_sigmoid



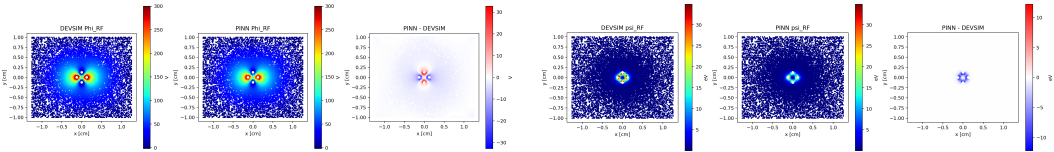
Final normalized loss 0.486; Phi MSE 3.1e+03; Psi MSE 65.6.

activation_sinusoidal



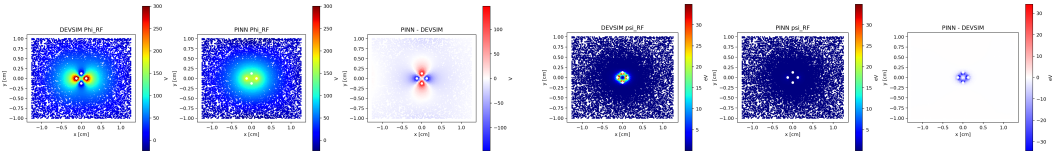
Best activation-only result; smooth derivatives help the PDE residual and gradient-derived Psi.

baseline



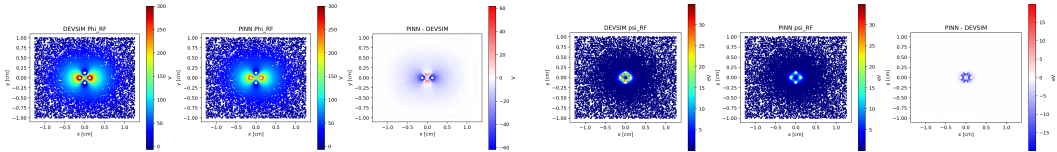
Reference run for all hyperparameter comparisons.

bc_weight_1



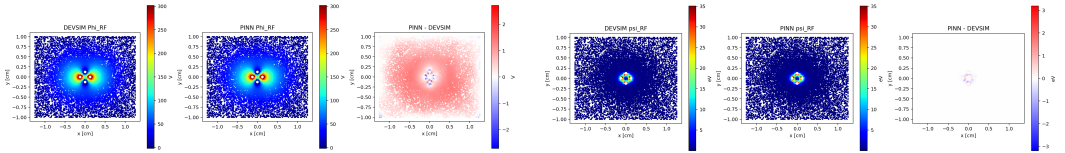
Final normalized loss 0.379; Phi MSE 2.44e+03; Psi MSE 63.1.

bc_weight_10



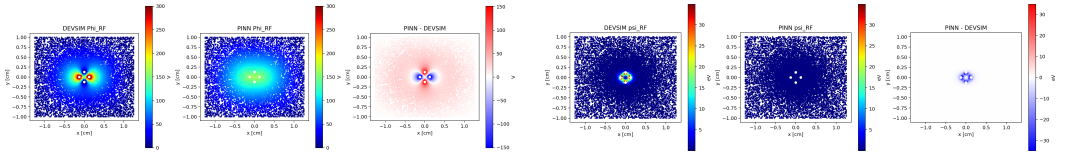
Final normalized loss 0.0814; Phi MSE 335; Psi MSE 19.3.

bc_weight_1000



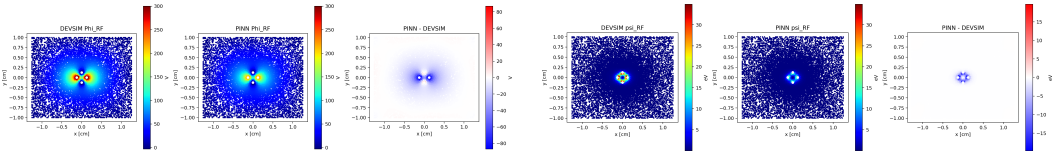
Best overall DEVSIM match; strong boundary enforcement did not destroy field accuracy.

depth_2



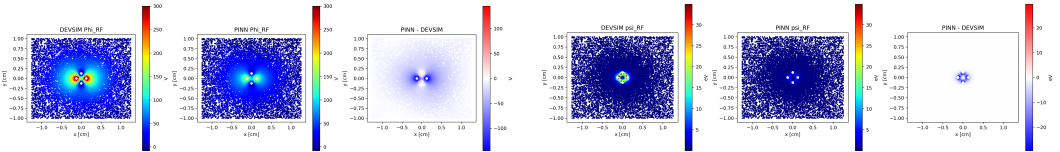
Final normalized loss 0.478; Phi MSE 2.95e+03; Psi MSE 65.4.

depth_6



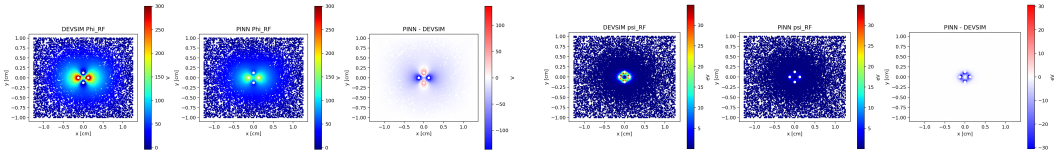
Final normalized loss 2.4; Phi MSE 886; Psi MSE 18.3.

depth_8



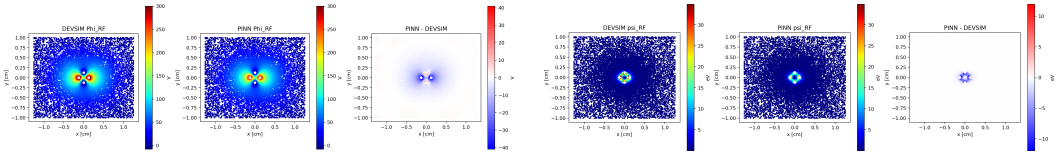
Final normalized loss 3.41; Phi MSE 2.81e+03; Psi MSE 44.4.

lr_1e-4



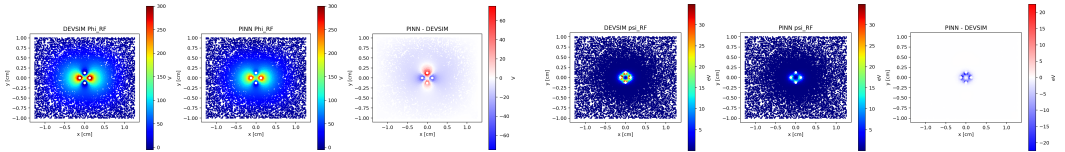
Final normalized loss 0.251; Phi MSE 1.95e+03; Psi MSE 50.2.

lr_3e-3



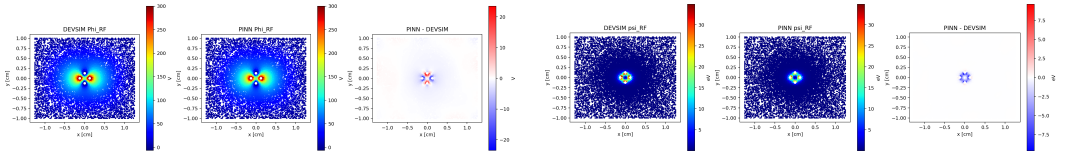
Final normalized loss 0.493; Phi MSE 142; Psi MSE 6.57.

lr_3e-4



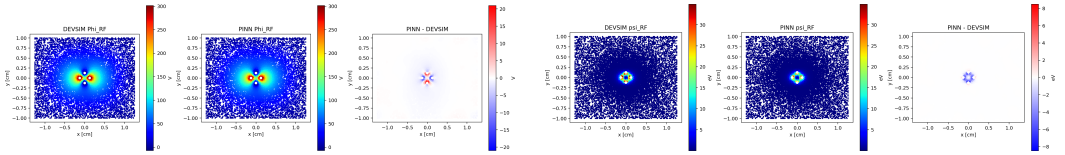
Final normalized loss 0.149; Phi MSE 358; Psi MSE 21.3.

optimizer_adamw



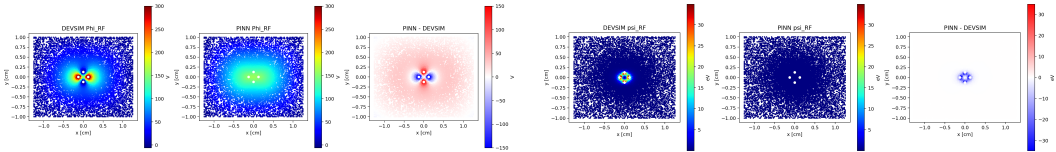
Final normalized loss 0.0103; Phi MSE 21.9; Psi MSE 3.33.

optimizer_rmsprop



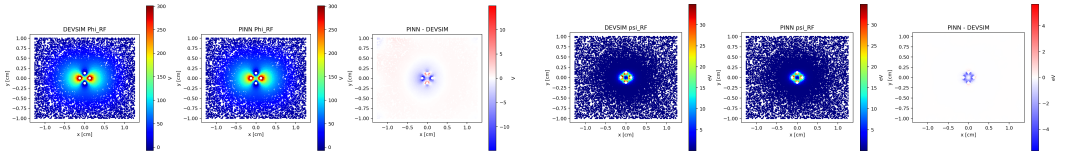
Final normalized loss 0.0145; Phi MSE 14.5; Psi MSE 2.38.

optimizer_sgd



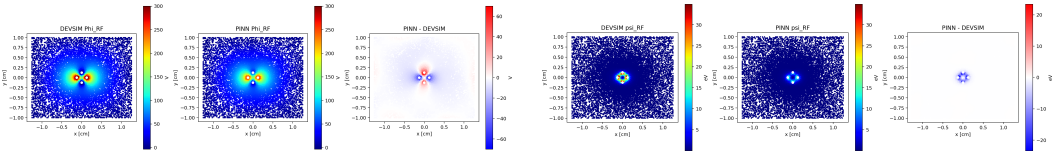
SGD plateaued; comparison confirms poor field accuracy for this training budget.

width_256



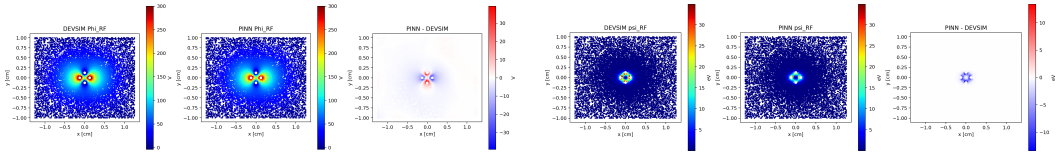
Best width result; wider representation improved physical field shape.

width_32



Final normalized loss 0.0767; Phi MSE 334; Psi MSE 22.5.

width_64



Final normalized loss 0.02; Phi MSE 69.2; Psi MSE 7.33.

Discussion Summary

Overall, normalized loss convergence and physical agreement with DEVSIM are related but not interchangeable. ReLU converged fastest by the normalized-loss metric, reaching 10^{-3} by step 2900 with final normalized loss 3.58e-08, but it produced the worst raw field mismatch (Phi MSE 7.2e+03, Psi MSE 3.97e+03); this makes it a negative control rather than a physically credible winner for a second-derivative PINN. Among credible runs, RMSprop reached normalized loss 10^{-2} first at step 7700, while BC weight 1000 reached 10^{-3} by step 1.1e+04 and gave the best total PINN–ground-truth MSE (Phi MSE 0.619, Psi MSE 0.0388). The next-best architecture-side cases were width 256 (Phi MSE 10.4, Psi MSE 0.955) and sinusoidal activation (Phi MSE 12.3, Psi MSE 1.75), which also had cleaner raw PINN–DEVSIM residual maps than the baseline. Depth increases, sigmoid, SGD, low boundary-condition weight, and the slowest learning-rate settings either failed to pass useful normalized-loss thresholds within the fixed training budget or retained broad residual structure in the raw error plots.

Variant	$t_{.1}$	$t_{.01}$	$\hat{\mathcal{L}}_f$	MSE_ϕ	MSE_Ψ
baseline	8000	>19.9k	0.0188	58.9	6.89
width_256	5800	1.03e+04	0.0685	10.4	0.955
width_32	1.7e+04	>19.9k	0.0767	334	22.5
width_64	1.17e+04	>19.9k	0.02	69.2	7.33
depth_2	>19.9k	>19.9k	0.478	2.95e+03	65.4
depth_6	1.09e+04	>19.9k	2.4	886	18.3
depth_8	>19.9k	>19.9k	3.41	2.81e+03	44.4
activation_relu	400	1000	3.58e-08	7.2e+03	3.97e+03
activation_sigmoid	>19.9k	>19.9k	0.486	3.1e+03	65.6
activation_sinusoidal	7700	1.2e+04	0.0089	12.3	1.75

Variant	$t_{.1}$	$t_{.01}$	$\hat{\mathcal{L}}_f$	MSE_ϕ	MSE_Ψ
bc_weight_1	>19.9k	>19.9k	0.379	2.44e+03	63.1
bc_weight_10	1.38e+04	>19.9k	0.0814	335	19.3
bc_weight_1000	5700	9000	0.00126	0.619	0.0388
optimizer_adamw	8700	1.43e+04	0.0103	21.9	3.33
optimizer_rmsprop	5100	7700	0.0145	14.5	2.38
optimizer_sgd	>19.9k	>19.9k	0.481	3.02e+03	65.5
lr_1e-4	>19.9k	>19.9k	0.251	1.95e+03	50.2
lr_3e-3	6000	>19.9k	0.493	142	6.57
lr_3e-4	1.61e+04	>19.9k	0.149	358	21.3

$t_{.1}$ and $t_{.01}$ are the first logged steps where normalized loss crossed 10^{-1} and 10^{-2}; >19.9k means not reached. Raw PINN–DEVSIM error maps for each row are shown in the preceding appendix slides.